

Syntax, Geometry, and Measurement:

*Towards a Unified Informational Foundation
for Logic and Relativity*

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Abstract

We present a unified foundational programme arguing that in logic, geometry, and physics alike, a common adjunction pattern separates cheap structural mathematics from costly semantic and ecological data. Four interconnected ideas are developed: a two-negation logic in which complement and opposition are distinct but adjoint involutions; an informational axiomatisation of relativity that carefully separates causal structure from metrical and dynamical assumptions; a derivation of measurement uncertainty from purely relativistic axioms, requiring no quantum mechanics; and a treatment of black holes as primitive curvature quasiparticles, from which mass emerges as a relational quantity. The thread connecting all four is the syntax-semantics adjunction identified by Lawvere, here traced through geometry and physics as well as logic. The framework dissolves several apparent puzzles — including dark matter and the measurement problem — by identifying them as artefacts of conflating structural mathematics with ecological data.

1. Introduction

Mathematics and physics have a complicated relationship with their own foundations. The twin revolutions of the early twentieth century — relativity and quantum mechanics — were each accompanied by foundational upheaval, but of rather different kinds, and the aftermath has been asymmetric in ways that are still not fully appreciated.

Quantum mechanics, despite its interpretational controversies, has attracted sustained foundational scrutiny. From von Neumann's algebraic formulation to the topos-theoretic approaches of Isham and Döring, and more recently to the reconstruction theorems of Hardy and Chiribella et al., there is a genuine and growing programme asking: *what minimal operational or informational axioms force quantum mechanics upon us?* Quantum information theory has become a mainstream field, with both theoretical depth and engineering applications in cryptography and computing.

Relativity, by contrast, has largely escaped this treatment. It is presented — even in advanced texts — as geometry plus physics: a beautiful mathematical framework whose physical content is taken largely for granted. The informational content of relativity, and the question of which parts of the standard presentation are genuinely mathematical versus contingent facts about our particular world, has received comparatively little attention. This paper is a contribution toward remedying that asymmetry.

Our central methodological principle can be stated simply. In any foundational framework — logical, geometric, or physical — there is a distinction between **structural mathematics** (the cheap, assumption-free skeleton of the theory) and **ecological data** (the additional input required to give that skeleton a specific instantiation). We argue that this distinction is mathematically precise: in each domain we consider, the relationship between these two levels takes the form of a **Galois connection** or adjunction. The failure of this adjunction to be an isomorphism is precisely the gap between syntax and semantics, between geometry and physics, between pure causality and measurement.

Before proceeding, it is worth briefly orienting the reader in the logical landscape. **Classical logic** treats negation as an involution — $\text{not}(\text{not}(A)) = A$ — and validates the law of excluded middle. **Intuitionistic logic**, developed by Brouwer and formalised by Heyting, relaxes this: a proposition need not be decidable, and double negation need not return you to where you started. The standard story presents this as a *relaxation* — the removal of an axiom. Our first idea proposes a different departure: rather than weakening the involution property, we retain it but *double it*, introducing two distinct involutions whose interaction encodes the syntax-semantics gap directly.

The paper develops this programme through four interconnected ideas, each inhabiting a different domain but each instantiating the same underlying pattern. We hope this is the beginning of a conversation rather than the end of one.

2. Two Negations

Logic begins with negation, and negation seems simple. In classical logic, *not* is an involution: $\text{not}(\text{not}(A)) = A$. Intuitionistic logic challenges this by observing that $\neg\neg A \rightarrow A$ fails in the absence of a completed proof. We wish to propose a different departure: rather than allowing negation to fail to be an involution, we introduce **two**

negations, each definitively an involution, but which are distinct and do not commute.

2.1 The two operators

Working in the setting of sets and subsets, the intended interpretations are:

not(A) — the *complement* of A: all elements outside A relative to a fixed universe. Purely syntactic and assumption-free.

neg(A) — the *antipodal set* of A: all elements maximally far from A, in some specified sense of opposition. This requires additional data — a metric, a convex hull, or an antipodal involution on the underlying space.

Both operators are order-reversing involutions. The Boolean case — a two-element universe — is a useful sanity check: here complement and opposition necessarily coincide, **not** and **neg** are the same operator, and we recover classical logic. In richer spaces they come apart.

2.2 The Galois connection axioms

We take as fundamental the following pair of axioms:

$$A \subseteq \text{not}(\text{neg}(A))$$

$$\text{neg}(\text{not}(A)) \subseteq A$$

The first says that nothing is its own opposite. The second says that the 'antipodally deep interior' of A is contained in A. Given that both operators are order-reversing involutions, these two axioms are not independent — each implies the other. Together they express precisely that **not** and **neg** form a **Galois connection**: **neg** is the right adjoint of **not** in the poset of subsets ordered by inclusion.

The compositions **not** ■ **neg** and **neg** ■ **not** become, respectively, a closure operator and an interior operator. The gap between them is the boundary — the region of genuine logical indeterminacy. The underlying structure resembles bi-Heyting algebras and orthocomplemented lattices, though our emphasis on two genuine involutions connected by an adjunction gives it a somewhat different character.

2.3 The Lawvere thread

Lawvere observed that syntax and semantics are adjoint functors. Our two negations instantiate this at the level of set-theoretic operations. **not** is the syntactic operator — free, structural, requiring no interpretation. **neg** is the semantic operator — requiring additional data, a way of specifying what counts as opposition in the world we are describing. The Galois connection between them is the syntax-semantics adjunction, made concrete and local.

The amount of data required to specify **neg** is precisely the amount of work required to provide a semantic interpretation for a given syntax. In the Boolean case this work vanishes: the two-element universe has only one possible involution, so the semantics is forced by the syntax, and logic is classical. This is the pattern we will see recur throughout the paper.

3. An Informational Axiomatisation of Relativity

The standard presentation of special relativity begins with light. We propose to strip light from the axioms entirely. We do not assume electromagnetism, photons, or Maxwell's equations. The speed of

light is not foundationally special because it is the speed of light — it is special because it is a maximum finite speed of signal propagation, and light happens to be a phenomenon in our world that saturates this maximum because photons happen to have zero rest mass. That is ecology, not mathematics.

3.1 The four levels

Level 0 — Causal Order. A set of events equipped with a partial order $A \leq B$ meaning 'event A can influence event B'. Purely combinatorial; no geometry, no metric, no notion of distance.

Level 1 — Cone Structure. A maximum finite speed of signal propagation, frame-independently defined. This adds a cone structure to the causal order — a boundary between events that can influence one another and those that cannot. No reference to what propagates at this speed. All genuinely relativistic content follows from Levels 0 and 1 alone.

Level 2 — Flat Metric (SR). The Minkowski metric imposed globally. The first genuinely ecological axiom — not derived from anything within the framework, simply stipulated. SR is sitting at Level 2 while pretending to be at Level 1: it makes a stronger and less justified assumption than GR while appearing more innocent. By fixing a flat metric without explanation, it does not even raise the question of where the geometry comes from.

Level 3 — Primitive Curvature Field (GR, decoupled). The metric becomes dynamical — replaced by a curvature tensor field treated as a primitive object, not necessarily sourced by anything. In standard GR the Einstein equations couple curvature to mass-energy:

$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

We propose to decouple them. The curvature field has its own autonomous dynamics — governed at minimum by the Bianchi identities, which are purely geometric — and the coupling to mass-energy is an additional ecological hypothesis, valid in our world but not foundationally required.

3.2 Dark matter reinterpreted

The dark matter puzzle arises because observed curvature in certain galactic contexts exceeds what can be accounted for by visible mass-energy via the Einstein equations. The proposed solution — invisible matter coupling gravitationally but not electromagnetically — is an ecological patch, invented to save the coupling equation. If the curvature field is primitive and not necessarily sourced by local mass-energy, there is no puzzle: the curvature field simply has a certain value in those regions. 'Dark matter' is an artefact of mistaking a contingent ecological hypothesis for a foundational law.

3.3 The parallel with Section 2

The causal/cone structure (Levels 0–1) corresponds to **not** — syntactic, assumption-free. The curvature field and its couplings correspond to **neg** — requiring additional data. The field equations are the Galois connection. The flat Minkowski metric of SR corresponds to the Boolean collapse: the special case where the adjunction degenerates, syntax and semantics coincide, and the geometry appears to need no explanation.

4. Relativistic Uncertainty

One of the most striking features of quantum mechanics is the Heisenberg uncertainty principle. We argue that this is, at least in part, an ecological misattribution. A fundamental and irreducible measurement uncertainty follows from the Level 1 axioms alone — from nothing more than the existence of a maximum finite speed of signal propagation and the finite time of flight of any measuring signal. No quantum mechanics is required. No Planck's constant appears.

4.1 The thought experiment

Consider two particles in a one-dimensional Euclidean universe. The observer sits at $x = 0$. The target is at unknown position $x = d > 0$ at $t = 0$, moving with unknown constant velocity v , where $|v| \leq c$. Its position at time t is $x(t) = d + vt$. The observer receives a spontaneous signal from the target at $t = 0$, then sends a test signal at $t = 0$ which returns at $t = T$.

4.2 Three events

Event A — target emits spontaneous signal at $t = -t_A$. The signal travels at c and arrives at $t = 0$:

$$t_A = d / (c + v)$$

Event B — test signal reaches target at $t = t_B$:

$$t_B = d / (c - v)$$

Event C — test signal returns at $t = T$, giving the fundamental relation:

$$d = T(c - v) / 2$$

4.3 The three positions

$$x_A = Tc(c - v) / 2(c + v)$$

$$x_B = Tc / 2$$

$$x_C = T(c + v) / 2$$

4.4 The central result

The position at Event B — the moment of measurement — is:

$$x_B = Tc/2$$

This is **independent of v** . Whatever the target's velocity, the position at the moment of reflection is exactly $Tc/2$. The observer knows T and c , and therefore knows x_B exactly.

The positions x_A and x_C , by contrast, depend on v and are genuinely uncertain. Since $v \in (-c, c)$: x_C ranges over $(0, Tc)$ — bounded above by the causal horizon — while x_A ranges over $(0, \infty)$ — unbounded, since the target could have been arbitrarily far away if moving toward the observer at near c .

Defining deviations from the measurement position:

$$\Delta x_A = x_B - x_A = Tcv / (c+v)$$

$$\Delta x_C = x_C - x_B = Tv / 2$$

Both vanish when $v = 0$, and their product is:

$$\Delta x_A \cdot \Delta x_C = T^2 cv^2 / 2(c+v)$$

To obtain a clean uncertainty relation — a product of variances bounded below — one would need a prior probability distribution over v and d . That prior is itself an ecological input, not forced by Level 1, and different priors yield different uncertainty bounds. What is forced by the axioms is the *structure* of the uncertainty: its dependence on v and T , and the exact determinacy of x_B .

4.5 The collapse structure

The observer, before sending the test signal, has genuine irreducible indeterminacy about the target's position. The test signal returns, and from the single observable T the observer deduces x_B exactly. The measurement *selects* a precise position from a continuum of possibilities.

This is the mathematical structure of **wavefunction collapse** in quantum mechanics — pre-measurement indeterminacy, exact determination at the moment of measurement, post-measurement uncertainty resuming — derived here from nothing but the Level 1 axioms. No Hilbert space, no operators, no Born rule, no Copenhagen interpretation.

Equally striking is the structure of the conjugate variables. In our massless universe there is no momentum in the mechanical sense $p = mv$. But velocity v and position play conjugate roles throughout: knowing T exactly determines x_B but leaves v entirely unconstrained. This is structurally identical to the complementarity of position and momentum in quantum mechanics, arising here from pure relativistic causality.

We note also a connection to Lamport's foundational work on distributed systems. His 1978 happens-before relation is a discrete, computational version of the Level 0 causal order. The uncertainty we have derived has a natural analogue in the clock skew and jitter that Lamport's framework is designed to manage. The connection deserves further investigation.

5. Curvature Quasiparticles

We now take the Level 3 programme to its logical extreme. If curvature is the primitive field, then the most concentrated, most coherent localisation of curvature is a **curvature singularity** — a point at which the curvature field diverges. We propose to invert the usual narrative: rather than treating black holes as exotic endpoints of stellar evolution, we treat them as the **primitive objects** of a Level 3 universe.

5.1 The general class

A **curvature quasiparticle** is any localised concentration of the curvature field — a region where curvature is significantly non-zero, surrounded by approximately flat regions. Singularities are the limiting, maximally coherent members of this class — the atoms of the Level 3 universe. Smooth curvature concentrations are the general case.

The dynamics is governed by the curvature field itself. Each quasiparticle deforms the geometry through which others move; those others follow geodesics in the deformed geometry, which in turn affects the curvature field. The interaction is fully self-referential — there is no fixed background, and no external force. The geometry is the interaction.

5.2 Identical singularities and the emergence of mass

For a system of n identical curvature singularities, there is no intrinsic scale — no quantity distinguishing one from another. Mass is not a meaningful parameter. The n -body problem for identical singularities is genuinely massless, governed purely by the geometry of relative configurations.

When singularities of different curvature strengths are considered, a meaningful comparison becomes possible: the ratio of curvature strengths. This ratio is the germ of what we call mass. Once a

reference singularity is chosen, every other singularity acquires a **mass parameter** measuring its curvature strength relative to the standard. The vacuum Einstein equations, with mass parameters appearing as integration constants, are then the appropriate dynamical equations for this more general case — but they are not foundational. They are the ecological specialisation of Level 3 dynamics to the case of non-identical singularities.

5.3 The Schwarzschild radius reinterpreted

The Schwarzschild radius is usually presented as:

$$r_s = 2GM / c^2$$

defined in terms of a body of mass M and the escape velocity equalling c . This presentation is ecologically loaded: mass M is a primitive input, escape velocity requires test particles, and G is a coupling constant that only makes sense once the coupling to mass-energy is established.

Stripped of this ecology, the Schwarzschild radius is simply a **geometric length scale intrinsic to the curvature singularity** — the scale at which the curvature becomes strong enough to close off causal contact with the exterior. It is a feature of the causal order, not of the ecology. For identical singularities, where mass is undefined, it is simply the natural unit of length set by the singularity's curvature profile.

5.4 Vacuum solutions as pure mathematics

General relativity has accumulated a rich catalogue of exact vacuum solutions — Schwarzschild, Kerr, Majumdar-Papapetrou, gravitational waves. In the standard presentation these are derived as special cases of the Einstein equations with $T_{\mu\nu} = 0$, and interpreted ecologically in terms of mass parameters and angular momenta.

Our framework invites us to read them differently: as **theorems about the Level 3 curvature field** — results about which configurations of curvature are consistent with the Bianchi identities and minimal dynamical assumptions. The mass parameters are not inputs but integration constants — geometric parameters characterising the singularity, which we can choose to interpret ecologically, or simply leave as geometric data.

Finally, the measurement uncertainty of Section 4 applies immediately to the n -body problem: two singularities attempting to determine each other's positions face exactly the uncertainty structure we computed. The instantaneous configuration of the system is not a well-defined observable. What is observable is the causal structure of signal exchanges — a Level 0 quantity. Whether this relativistic indeterminacy is a precursor to the quantum gravitational indeterminacy of loop quantum gravity is a question we leave open as a potentially fruitful direction.

6. Synthesis and Open Questions

We have travelled through four domains — logic, geometry, relativistic physics, and the foundations of measurement — and found the same pattern at work in each. The unified table is as follows:

Domain — Structural — Ecological — Adjunction

Logic: Complement (not) | Antipodal map (neg) | Galois connection
Geometry: Causal/cone structure | Curvature, metric | Field equations

Measurement: Signal timing | Prior over v , d |
Uncertainty relation
Gravitation: Curvature singularities | Mass, matter
content | Vacuum equations

In every case the structural level is free; the ecological level costs something; and the relationship between the two is a Galois connection. The classical, flat, or Boolean special case is always the one where the adjunction degenerates to an isomorphism — concealing rather than eliminating the gap.

6.1 The methodological principle

In any foundational framework, the distinction between structural mathematics and ecological data is not merely philosophical but mathematically precise — it is the distinction between the two sides of a Galois connection. Conflating them leads to confusion; separating them is clarifying. Dark matter is a puzzle only if you conflate the curvature field with its mass-energy source. Quantum uncertainty feels mysterious only if you attribute it entirely to quantum mechanics. The incompatibility of QM and GR feels fundamental only if both are taken as foundational without examining their ecological assumptions.

6.2 The Lawvere thread

Underlying the entire paper is Lawvere's observation that syntax and semantics are adjoint functors. What we have done is trace the consequences of taking this observation seriously not just in logic but in geometry and physics. The antipodal map, the curvature field, the prior distribution, the mass parameter — all are instances of the same thing: the semantic data that gives a syntactic skeleton its meaning in a particular world. Whether the specific structures developed here fit cleanly into the categorical framework of Lawvere theories and toposes is a question we consider highly promising.

6.3 On loop quantum gravity

Loop quantum gravity begins from the observation that the gravitational field is itself a connection, and attempts to quantise it without presupposing a fixed background metric. This is philosophically aligned with our Level 3 framework. The spin networks and spin foams of LQG are, in a sense, discrete Level 0 causal structures. The relativistic uncertainty of Section 4 — not quantum but causal — suggests that the discrete granular geometry of LQG may have a classical relativistic precursor. Whether the quantisation of area and volume in LQG is partially anticipated by Level 1 causal structure imposing a minimum resolvable scale on measurement is an open and, we think, genuinely interesting question.

6.4 On quantum mechanics

We have been careful not to claim that quantum mechanics reduces to our framework. The claim is more modest: some of the structure usually attributed specifically to quantum mechanics — collapse, conjugacy of position and momentum, irreducibility of uncertainty — is already present at the purely relativistic level. This suggests that QM and relativity may share a deeper common foundation in the informational structure of measurement, and that the apparent tension between them may be partly an artefact of each carrying ecological assumptions at different levels of the hierarchy. The reconstruction theorems for QM provide a template for the work that remains to be done for relativity.

6.5 On the two-negation logic

The specific combination of two genuine involutions connected by a Galois connection does not seem to correspond exactly to any well-studied structure in the literature, though connections to bi-Heyting algebras and orthocomplemented lattices are clear. A full algebraic and categorical analysis — the appropriate notion of morphism, the free algebra on generators, the relationship to topos theory — remains to be carried out. The proof-theoretic question is also open: what sequent calculus captures the axioms of Section 2, and what does proof by **not**-elimination mean compared to proof by **neg**-elimination?

6.6 Closing remark

The ideas in this paper grew out of a simple observation: that the distinction between syntax and semantics, familiar from logic, reappears in geometry and physics in the guise of the distinction between structure and ecology. We offer the framework in the spirit in which it was developed — as the beginning of a conversation, conducted in the belief that the right questions are at least as valuable as the right answers.

7. Ecological Contingency, World-Building, and the Cost of a Fork

The framework developed in the preceding sections has a natural extension that we did not anticipate at the outset, but which emerges from it with some inevitability. We have argued that ecological data is additional input not forced by the structural mathematics, connected to that skeleton by a Galois connection. A natural question, which we have so far left unasked, is: *what does it cost to realise a particular ecological choice?* The answer is both quantifiable and deeply connected to the uncertainty structure of Section 4.

7.1 Landauer's principle and its generalisation

Landauer's principle asserts that the erasure of one bit of information has a minimum thermodynamic cost of $kT \ln 2$ in energy. This establishes a direct bridge between information-theoretic operations and physical costs — between the logical and the thermodynamic.

Vaccarino and Barnett showed that Landauer's principle is not special to the energy-entropy conjugate pair. It generalises to *any* pair of conjugate variables — any two quantities satisfying an uncertainty relation. Wherever there is a conjugate pair, there is a meaningful notion of information-theoretic cost denominated in those variables. The energy-entropy accounting of thermodynamics is one instance of a universal principle: **conjugate pairs are accounting currencies**.

This result transforms the landscape of our framework. In Section 4 we derived an uncertainty relation from purely relativistic, Level 1 axioms — between position and velocity, with no quantum mechanics and no mass. The Vaccarino-Barnett result then applies immediately: this relativistic conjugate pair is already a valid accounting currency. The causal structure alone equips us with a meaningful cost measure, before any ecology is introduced.

7.2 The many-worlds perspective

The structural mathematics — causal order, cone structure, primitive curvature field — is the common substrate shared by all possible worlds. The ecological data is what individuates one world

from another: the particular antipodal map, metric, matter content, and coupling constants. The space of all consistent ecological assignments to a given structural skeleton is the space of all possible worlds compatible with that skeleton.

Our world is one point in this space. The structural mathematics is indifferent between the points. The question 'why this ecology?' is not one the mathematics can or should answer — and this **silence is not a deficiency but a feature** of the framework, exactly as the silence of classical logic about which semantic interpretation to choose is a feature of its generality rather than a gap in its foundations.

7.3 Forking the world locally

Consider an agent within a Level 3 universe wishing to realise, locally within some bounded region of spacetime, a *different* ecological choice from the one prevailing globally. They wish to fork the world: to create a local patch in which the curvature field, matter content, or effective metric takes values different from those of the surrounding universe.

The causal structure constrains how this can be done. Signals propagate at most at speed c . The agent cannot simultaneously freeze the state of the universe and rearrange it; information about their rearrangement propagates outward and the universe responds. To successfully fork, the agent must *outcompute the universe* within their target region — rearrange the local state faster than the causal structure can propagate the consequences outward.

The cost of this operation is the cost of suppressing or redirecting causal propagation — of pausing the relevant degrees of freedom while the fork is instantiated. This cost is denominated in the conjugate variable to whatever quantity is being rearranged, by the Vaccarino-Barnett principle. The blockchain analogy is precise: a fork requires a shared ledger (the structural skeleton), a local rearrangement of state (a new ecological choice), sufficient computational work to make the fork persist, and consistency with the global causal structure. The cost is real, quantifiable, and denominated in the canonical variable conjugate to the quantity being forked.

7.4 Which numeraires are meaningful

The Vaccarino-Barnett result *qualifies* which quantities can serve as meaningful numeraires for world-building. Not every pair of quantities is a valid accounting currency — only conjugate pairs, those satisfying an uncertainty relation, have the right structure to support a Landauer-type cost theorem.

This gives our framework a precise answer to which variables matter for ecology. Conjugate pairs are generated by the adjunction structure: the Galois connection between **not** and **neg**, instantiated at each level of the hierarchy, generates pairs of quantities optimally related without being identified. The uncertainty relation is the observable shadow of the adjunction in the space of physical quantities. The hierarchy is then:

Adjunction $\rightarrow$ Conjugate pairs $\rightarrow$ Uncertainty
relations
 $\rightarrow$ Valid numeraires $\rightarrow$ Cost of ecological
fork

7.5 The conjugate to curvature

A natural and important open question emerges immediately: *what is the conjugate to curvature in a genuinely Level 3 framework?* In canonical quantum gravity, the conjugate to the spatial metric g_{ij} is the extrinsic curvature K_{ij} — the gravitational momentum at the heart of the Wheeler-DeWitt equation. But this derivation leans on the Einstein-Hilbert action, which is itself an ecological input in our framework.

The question is therefore genuinely open: what is the conjugate to curvature before any ecological action principle is chosen, derived from the Bianchi identities and the Level 1 causal structure alone? Whatever lives there will determine the natural accounting currency for curvature-forking — for locally instantiating a different gravitational ecology — and will likely connect the causal structure, the Bianchi identities, and the Vaccarino-Barnett principle in a way not yet fully understood. We regard this as the most important open question generated by the present framework.

7.6 The cost of classical logic

We close with an observation connecting back to Section 2. Classical logic is the special case where the adjunction between **not** and **neg** degenerates to an isomorphism — the semantic data is forced by the syntactic structure, and no ecological choice remains. The Boolean world costs *nothing* to maintain: there is no fork, because there is no freedom.

Conversely, the richer the adjunction — the larger the space of ecological choices compatible with a given structural skeleton — the greater the potential cost of any particular fork, and the more meaningful the accounting. A world with a trivial adjunction is a world with no room for world-building. A world with a rich adjunction is a world in which agents can, at a cost, locally realise alternative ecologies — can, in a precise sense, *do physics*.

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